



CBRS-X : HACKATHON PITCH MASTER BRIEFING & TEAM BATTLE PLAN

Enterprise Tactical VR & WebGL Chemical, Biological & Radiological Emergency Training with Deterministic Audit Engine

TEAM LEAD

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01 PROJECT OVERVIEW & MISSION NARRATIVE



THE REAL-WORLD LIFE-OR-DEATH CRISIS

In real-world CBRN emergencies, a single procedural error (e.g. stepping into a hot zone without positive-pressure SCBA or misreading gas PPM gradients) leads to fatal casualties.

Cost Barrier: Physical drills require single-use Level-A hazmat suits costing ₹50,000–₹1,00,000 each and cannot be repeated safely.



THE CBRS-X PARADIGM SHIFT

Zero Physical Risk: Responders navigate an interactive 3D digital twin of *Storage Bay 03* in VR or browser WebGL on standard laptops.

Deterministic Auditing: Evaluates every timestamp, gas PPM threshold, and suit seal objectively against official NDRF Standard Operating Procedures (SOPs).

1. DETECT (PID Gas Sensor)



2. PROTECT (Level-A PPE & SCBA)



3. CONTAIN (Magnetic Clamp Seal)



4. EVACUATE (Casualty Triage)



5. DECONTAMINATE (Shower Archway)

SYSTEM TIER	TECHNOLOGIES USED	PLAIN ENGLISH FUNCTION
1. Frontline Simulation	Unity 2022.3 LTS, WebGL / WASM, OpenXR, Three.js	The interactive 3D virtual environment (Bay 03) where trainees move, check gas detectors, patch leaking drums, and take decon showers.
2. Trainee Web Station	React 18, Vite 5	The browser portal where trainees log in, launch the simulation window, view live mission objectives, and read HUD metrics.
3. Backend & Audit Core	Java 17, Spring Boot 3.2.5, STOMP WebSockets	The central engine that ingests event telemetry, calculates 100-point scores, deducts penalty points, streams live radar data, and generates verified PDF certificates.
4. Command & Storage	React 18 Glassmorphism, PostgreSQL 15	The supervisor dashboard with a live 2D/3D radar showing responder position at 60Hz, mission replay controls, and database audit logs.

**SUB-50MS TELEMETRY**

STOMP WebSockets stream responder actions and gas levels at 60Hz.

**SHA-256 CERTIFICATES**

Tamper-evident cryptographic digest embedded in generated completion PDFs.

**INGRESS RATE LIMITING**

Bounded LRU Sliding-Window Rate Limiter tested for 10,000 concurrent IPs.

1. DETERMINISTIC AUDIT CORE

Not just a game; it is a millisecond-accurate compliance engine evaluating real NDRF protocols without human bias.

2. ZERO HARDWARE BARRIER

Runs directly inside Google Chrome or Microsoft Edge on standard budget laptops via WebGL / WebAssembly.

3. DUAL-SCREEN COMMAND SYNC

Trainee first-person view is synchronized in real time with the Instructor Tactical Radar HUD.

**5-MINUTE PITCH TIMING ROADMAP:**

[0:00-0:45] The Hook & Problem → [0:45-1:30] The Solution (CBRS-X) → [1:30-3:00] Live Dual-Screen Demo → [3:00-3:45] Tech Architecture & Test Stats → [3:45-4:30] ROI & Market → [4:30-5:00] Closing & Q&A

SPEAKER 1: MISSION COMMANDER / PITCH LEAD

Hook, Problem Statement & Solution

"Judges, in a CBRN disaster, there is zero room for error. A single mistake—stepping into a hot zone 5 seconds before your SCBA seals or misreading a gas detector by 10 PPM—is fatal. Yet live physical drills cost over ₹50,000 per suit in consumables and cannot be repeated safely. Our solution is CBRS-X—an enterprise tactical simulation and deterministic auditing platform that brings zero-risk, high-fidelity emergency training directly into standard web browsers."

SPEAKER 2: TECHNICAL ARCHITECT / DEMO DRIVER

Live Dual-Screen Demo & Telemetry

"On the left is our Trainee WebGL Station running in the browser; on the right is the Instructor Tactical Command Center. As our responder navigates Storage Bay 03, every movement is streamed at 60Hz over WebSockets. Watch what happens when our responder approaches the toxic plume without donning Level-A PPE: the backend instantly flags an SOP safety breach, logs a penalty, and alerts the instructor. Once contained and decontaminated, the system dynamically issues a tamper-evident SHA-256 verified PDF certificate."

SPEAKER 3: BACKEND & SECURITY SPECIALIST

Tech Architecture, Scoring & Test Coverage

"Under the hood, CBRS-X is built on an enterprise-grade stack: Unity 2022.3 compiled to WebAssembly with Brotli compression for 65% smaller bundle sizes; Spring Boot 3.2.5 microservices handling deterministic state machines; and a Bounded LRU Rate Limiter handling 10,000 concurrent IPs. Our backend is validated with 46 out of 46 unit and integration tests passed."

SPEAKER 4: PRODUCT VISIONARY / IMPACT LEAD

Real-World ROI, Field Box & Market Reach

"The real-world impact is transformative: 100% physical risk elimination and a 90%+ cost reduction for 16 NDRF battalions, state SDRF units, and energy giants like IOCL, ONGC, and GAIL. Through our Docker 'Field Box' architecture, CBRS-X runs completely offline in remote border bases on a single local laptop Wi-Fi hotspot with zero internet required."

3-STEP LIVE DEMO CHOREOGRAPHY

- **Step 1 (Movement & Detection):** Move responder $\rightarrow$ show radar blip moving $\rightarrow$ show PID gas detector PPM value rising.
- **Step 2 (Penalty Trigger):** Approach toxic plume without PPE $\rightarrow$ point to instructor feed showing `entered_hazard_zone_without_ppe` penalty.
- **Step 3 (Containment & Cert):** Don Level-A suit $\rightarrow$ seal leaking drum $\rightarrow$ pass through decon arch $\rightarrow$ download SHA-256 PDF certificate.

TROUBLESHOOTING & FALLBACKS

- **Pre-load WebGL Tab:** Spawn into Bay 03 *before* walking up to the judges to avoid loading lag.
- **Local Docker Fallback:** Keep `docker-compose` up running locally on `localhost:80` in case venue Wi-Fi drops.
- **If WebGL frame drops:** Explain: *"This is running on standard integrated graphics without a GPU, proving zero-install accessibility."*

? "How is this different from existing VR games or training simulators?"

 **Answer:** "Most VR simulators are graphical walkthroughs with subjective grading. CBRS-X is a **deterministic auditing engine**. Every gas PPM threshold, suit seal duration, and decontamination dwell time is evaluated with millisecond accuracy against official NDRF SOPs, producing tamper-evident SHA-256 verified credentials."

? "Can low-end government computers in remote training camps run this?"

 **Answer:** "Yes. By utilizing Unity WebGL with Brotli compression and LOD shaders, it runs inside standard Chrome on basic laptops. Furthermore, our Docker 'Field Box' allows the entire backend and DB to run offline on a single rugged laptop acting as a local Wi-Fi hotspot."

? "How do you ensure certificates cannot be forged or faked?"

 **Answer:** "Every certificate embeds an immutable **SHA-256 cryptographic digest** computed from the session ID, trainee ID, final score, and millisecond timestamps. Training academies can verify authenticity offline simply by recomputing the hash."

? "How difficult is it to add new disaster scenarios (e.g., Radiation or Biohazards)?"

 **Answer:** "Our backend scoring engine is polymorphic. Hazard thresholds, detector types, and penalty weights are defined in modular JSON state machines. New scenarios (like radiological dirty bombs or bio-containment) plug in without refactoring the core pipeline."

